

Logistic Classifier — Notes Summary

Perceptron vs Logistic Regression

Perceptron (Discrete)

Logistic Regression (Continuous)

Activation: Step function-> perceptron

Sigmoid function->Logistic

Training the Logistic Regression

1. Start with random line (w_1, w_2, b)
2. Repeat:
 - Select a point
 - Compute error
 - Update w_1, w_2, b (fine-tune)

3. Until stop condition satisfied

Stop conditions: number of epochs reached, OR minimum error reached

Error types: sum absolute, mean absolute, sum square, mean square, RMSE, **log loss** (used in sigmoid/logistic systems)

Computing the Error

Characteristics of a good error function:

- Score is high when prediction is true (matches label)
- Score is low when prediction is false (doesn't match label)

Absolute error: $|y - \hat{y}|$ — large when far from label, small when close

Square error: $(y - \hat{y})^2$

Log Loss

- Let **p** = probability point is positive (+ve)
- Let **h** = probability point is negative (-ve) = $1 - p$

Error rule:

- If $y = 1$: error = $1 - \hat{y}$
- If $y = 0$: error = $\hat{y}$

Since independent probabilities multiply: $P(A \cup B) = P(A) \cdot P(B) \rightarrow \ln(P(A \cup B)) = \ln(P(A)) + \ln(P(B))$

So instead of multiplying, we sum the logs (negative log of probability of correct class):

Log loss formula:

- If label = 0: log loss = $-\ln(1 - \hat{y})$
- If label = 1: log loss = $-\ln(\hat{y})$

Combined formula:

Log loss = $-y \cdot \ln(\hat{y}) - (1 - y) \cdot \ln(1 - \hat{y})$

Total classifier error = sum of individual log losses

The Logistic Trick (Gradient Update Rule)

For a point with learning rate η :

- $a' = a + \eta(y - \hat{y})x_1$
- $b' = b + \eta(y - \hat{y})x_2$
- $c' = c + \eta(y - \hat{y})$